MIL-P-43772A
23 October 1987
SUPERSEDING
MIL-P-43772
3 May 1972

MILITARY SPECIFICATION

PANEL, TYPE II MODULAR PLATFORM, AIRDROP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of balsa core, aluminum faced, structural sandwich composite panel.

2. APPLICABLE DOCUMENTS

2.1 *Government documents.*

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

MMM-A-132	- Adhesives, Heat Resistant, Airframe Structural, Metal-to-Metal
MMM-A-188	- Adhesive; Urea-Resin-Type (Liquid and Powder)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-S-7998 - Sandwich Construction Core Material, Balsa Wood
MIL-A-9067 - Adhesive Bonding, Process and Inspection, Requirements For

STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-401 - Sandwich Construction and Core Materials; General Test Methods

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

11-1-318 - Platform, Air Delivery, Modular, (Dual Rail System) Type II; Panel, Sandwich, Balsa Core, 4 Ft. Assembly

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials and components. The materials, components, and supplementary processes involved in fabrication of the panel shall conform to the details, requirements, and dimensions shown or specified on Drawing 11-1-318 and details relating thereto, the bonding process specification (see 3.3.2.1), and as specified herein.

3.2.1 Materials.

3.2.1.1 Balsa wood. The balsa wood planks (individual pieces) used in construction of the panel core shall conform to MIL-S-7998 except that the paragraphs on "Density," "Corcho or low-density balsa," and "Presence of low-density (corcho) material" shall not apply.

3.2.1.2 Adhesive. Adhesive other than those specified on Drawing 11-1-318 shall not be used without prior approval of the contracting activity. Any request for deviation shall be accompanied by data proving conformance of the proposed alternate adhesive to the performance requirements of MMM-A-132 or MMM-A-188, as applicable.

3.3 Construction. Construction shall conform to the requirements of Drawing 11-1-318 and details relating thereto and as specified herein.

3.3.1 Blocks and cores. Blocks and cores shall be formed [illegible] planks conforming to 3.2.1.1. The average density of blocks or core[illegible] shall not exceed 10 pounds per cubic foot.

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3.3.1.1 Dimensions. Dimensions of the blocks and rough cores shall be as specified in table I. Finished cores shall conform to the assembly requirements specified on Drawing 11-1-318. Tolerances for length, width, and thickness shall conform to the requirements of the particular bonding process used.

TABLE I. Dimensions of blocks and rough cores

	Length	Width	Thickness
Blocks	Random - 1 ft minimum (no butt joints)	Contractor's option (normal mill maximum - 60 in.)	Contractor's option (normal mill maximum - 15 in.)
Cores, rough	Contractor's option (normal mill maximum - 48 in.)	Contractor's option (normal mill maximum - 24 in.)	Contractor's option (no butt joints)

3.3.1.2 Bonding. Area of effective bond (wood tearing in shear) at the faying surfaces of planks or block slices when formed into cores shall be not less than 75 percent when tested as specified in table II.

3.3.2 Panel assembly.

3.3.2.1 Bonding. Bonding of the panel assembly (core to core, core to metal, and metal to metal) shall be in accordance with a complete detailed process specification prepared by the contractor in accordance with MIL-A-9067, as modified below. (Whenever the term "process specification" occurs in this document, it shall be interpreted as the contractor's process specification.)

a. The process specification shall be made available for review by, and approval of, the contracting activity concurrent with first article inspection and approval (see 4.3 and 6.2). In the event of first article disapproval, approval of the process specification shall be deferred pending any required modifications thereof.

b. Preparation of faying surfaces - The process specification shall contain specific provisions and tests to ensure that, immediately prior to bonding, the faying metal surfaces are clean and of a chemical nature such as

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to provide a permanent, high strength bond. Assessment of the suitability of the surfaces for bonding shall not depend solely on a water-film test, as such testing does not guarantee the absence of cleaning solution films and undesirable sealing of etched surfaces. The suitability of the adhesive supplier's recommended surface treatment, if used, shall be verified by the panel supplier. Elapsed time between treatment and bonding shall be kept to an absolute minimum.

c. Application of adhesive - The process specification shall include specific provisions and tests to ensure that the amount of adhesive used, application system, and final pressing method will provide proper flow, entire wetting of faying surfaces, and adequate final bond-line thickness.

d. Prefitting of parts - The process specification shall include specific procedures and provisions for visual inspection to ensure that core sections, when installed in the panel frame, are closely butted one to the other and to the faying frame surfaces as specified and shown on Drawing 11-1-318, and that glue lines at such points are visible and continuous.

e. Nondestructive tests - The process specification shall include a procedure for nondestructive testing capable of detecting, as a minimum, 1 1/2 inch average diameter near side unbonds between the facing sheets and the adhesive (see 6.5). The procedure shall include detailed descriptions of the following:

(1) System instrumentation, including operation and maintenance.

(2) Instrument calibration standards, including correlation test panels with artificially induced unbonds and missing adhesive. Mechanical contact in way of defects in test panels shall be good.

(3) Qualification of operating personnel.

f. Destructive inspection - Except as may be required by the contracting activity for determination of cause of any first article failure, destructive inspection of complete panels shall not be required.

g. Changes - The contracting activity shall be advised of any proposed changes to the approved process specification during production. Any change considered of sufficient significance by the contracting activity to warrant review and approval thereby shall be so processed (see 4.4.5).

3.3.2.2 Proofload and bond quality. The unpainted panel assembly shall meet the requirements listed below when tested as specified in 4.4.5.

a. No visible permanent deformation of the side or end channels.

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b. No visible wrinkling, buckling, dimpling, or shear crimping of the facing sheets.

c. No shearing or loosening of any rivets.

d. No separation of facing sheets from side or end channels.

e. No detectable unbonded area within 4 inches of facing sheet edges.

f. No two detectable unbonded areas closer to each other than 6 inches.

g. No detectable unbonded area in excess of 1 1/2 inches average diameter.

h. Four or less detectable unbonded areas in either facing sheet bond.

3.4 *Government-loaned equipment.* The items listed below are required for panel proofload testing (see 4.4.5 and figure 2) and may be obtained from the contracting activity on a loan basis (see 6.2).

Item	Part No.	NSN
Side rail	11-1-321-1	1670-00-893-1625
Clevis	MS70085-1	1670-00-893-1631

3.5 *Workmanship.* The finished panel shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 *Responsibility for inspection.* Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.4.5. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection. In addition to the detailed provisions of table II, inspection shall be made at appropriate stages of fabrication to determine conformance to the requirements of this document and the process specification, including the dimensional requirements. The Government reserves the right to exclude from consideration for acceptance any materials for which in-process inspection has indicated nonconformance.

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TABLE II. In-process inspection of blocks and cores

Inspection of	Characteristic	Requirement para.	Sample unit	Insp. procedure	Insp. plan	Results reported as
Blocks	Length	3.3.1.1	1 block	Measure	1/	Pass or fail
	Butt joints			Visual	1/	Pass or fail
Rough cores	Butt joints	3.3.1.1	1 core	Visual	1/	Pass or fail
	Bonding	3.3.1.2	Section of rough core	Core shear MIL-STD-401 2/	1/	Percent bond area
Finished cores	Density	3.3.1	1 core	Compute $\frac{W}{V}$	1/	Pounds per cubic foot

1/ In accordance with the process specification.

2/ Stress/strain data not required.

4.4.2.1 Screening. Both faces of each panel shall be examined, prior to painting, for any evidence of buckling, bulges, or wrinkles. Any evidence of such defects shall be cause for rejection of the panel.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table III. The lot size shall be expressed in units of completely assembled airdrop panels. The sample unit shall be one panel. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.0 for major defects and 6.5 for total (combined major and minor) defects.

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TABLE III. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Final paint	Color not as specified		201
	Tacky (not dry to touch)		202
	Color separation		203
	Incomplete		204
	Blistered, peeling, flaking, or otherwise not adherent		205
	Embedded dirt or other foreign matter		206
	Creeps, wrinkles, pits, sags, streaks, or runs		207
Construction and workmanship	Any component missing, cracked, fractured, misaligned, or malformed (except as otherwise classified herein)	101	
	Any component not assembled or joined as specified	102	
	Any gap in facing sheet perimeter seal, including protruding gauze	103	
	Adhesive overrun		208
	Any sharp edges, burrs, or metal slivers		209
	Any machine screw or rivet missing, loose, or of wrong type	104	
	One rivet not properly set (i.e., head cracked, cocked, ringed, or not flush)		210
	More than one rivet not properly set	105	
	Facing sheet damage from riveting equipment or flush head shaver		211
Marking	Missing, incomplete, not legible, or otherwise not as specified		212

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4.4.4 End item dimensional examination. The end items shall be examined for the dimensions on Drawing 11-1-318. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of completely assembled airdrop panels. The sample unit shall be one panel. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 10.0.

4.4.5 End item testing. The end items shall be tested in sequence as specified in 4.5 to determine compliance with the proofload and bond quality requirements of 3.3.2.2. For the first 50 panels produced, the inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 1.5. For succeeding production, the inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. The lot size shall be expressed in units of completely assembled airdrop panels. The sample unit shall be one panel. In the event of rejection of any lot, or if required by the contracting activity for assessment of changes to the process specification (see 3.3.2.1(g)), the first 10 panels produced following such rejection or change shall be tested as specified herein.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Material	Not style and grade specified Any component missing, damaged, defective, or unduly dirty or stained
Workmanship	Panel not fitting snugly in container Loose strapping or inadequate stapling
Marking	Omitted; incorrect; illegible or incomplete legend; incorrect location, color, or size

4.5 Methods of inspection.

4.5.1 Test No. 1. The panel, supports, load pads, and load shall be arranged as shown on figure 1 on any suitable solid surface such as a concrete floor. The specified load of 9,000 pounds (static) shall be applied for a period of not less than 10 seconds in the direction shown on figure 1. The load shall then be removed and the panel examined for conformance to

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requirements (a) through (d) listed in 3.3.2.2. If no defects are found, the panel shall be turned over and the test repeated on the opposite face. If no defects are found, the panel shall then be tested as specified in 4.5.2.

4.5.2 Test No. 2. The panel, side rail, clevises, and pipe or rod shall be assembled as shown on figure 2. The assembly shall then be secured to any suitable solid surface, such as a concrete floor or metal fixture, by means of any suitable holding device positioned as shown on figure 2. Orientation of the panel relative to a horizontal plane shall be optional. The specified load of 9,000 pounds (static) shall be applied for a period of not less than 10 seconds in the direction shown on figure 2. The load shall then be removed and the panel examined for conformance to requirements (a) through (d) listed in 3.3.2.2. If no defects are found, the assembly shall be disassembled, the panel turned over and reassembled, and the test repeated. If no defects are found, the test shall be repeated on the remaining two opposite outboard edges. If no defects are found, the panel shall then be tested as specified in 4.5.3.

4.5.3 Test No. 3. The panel shall be tested for compliance with 3.3.2.2 (e) through (h) in accordance with the process specification nondestructive test procedure (see 3.3.2.1 (e)).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.1.1 Level A packing. Each panel shall be packed in a snug-fitting, wood-cleated or nailed wood shipping container conforming to overseas type, style A or J, grade A or B, type 1 load of PPP-B-601, or class 2, style 2 or 4, grade A or B, type 1 load of PPP-B-621. Each container shall be provided with skids as specified in the applicable container document. The inside dimensions of the shipping container shall be approximately 105 inches in length, 49 inches in width, and 4 inches in depth. Approximate dimensions are furnished as a guide only. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container document.

5.1.2 Level B packing. Each panel shall be packed in a snug-fitting, fiberboard shipping container conforming to style FTC, grade V2s of PPP-B-636. The inside dimensions of the shipping container shall be approximately 105 inches in length, 49 inches in width, and 4 inches in depth. Approximate dimensions are furnished as a guide only. The size and weight limitations for the fiberboard box shall be waived. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-636. Eight reinforcing bands shall be required. Two bands shall first be applied lengthwise and then six bands shall be applied girthwise.

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5.1.3 Commercial packing. Panels shall be packed in accordance with ASTM D 3951.

5.2 Marking. In addition to any special markings required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The airdrop panel is the major component of a modular airdrop platform used as a base for supporting cargo loads during transportation, extraction from aircraft, and ground impact. Platforms 8, 12, 16, 20, 24, and 28 feet in length can be assembled in the field by utilizing the appropriate components as shown on Drawing 11-1-317. The platform is used for airdrop from C-130, C-141, and C-5A aircraft equipped with platform restraint rail systems.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 3.3.2.1(a), 4.3, and 6.3).
c. Provisions for Government loaned equipment (see 3.4).
d. Selection of applicable level of packing (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Contract data requirements. Data conforming to Data Item Description DI-R-1725 will usually be required for delivery in connection with this document (see 6.2(c)). When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.5 Nondestructive tests. The following nondestructive test methods are suggested for consideration in development of the process specification (see 3.3.2.1(e)).

a. Contact impedance preliminary scan, followed by high frequency pulse-echo resolution of suspect areas.

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b. Pulsed eddy sonic.

c. Thermographic.

6.6 Subject term (key word) listing.

Airdrop system
Panel, airdrop
Platform, airdrop

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - AS
Air Force - 11

Review activities:

Army - AV, TS
Air Force - 82, 99

User activity:

Navy - MC

Preparing activity:

Army - GL

Project No. 1670-0754

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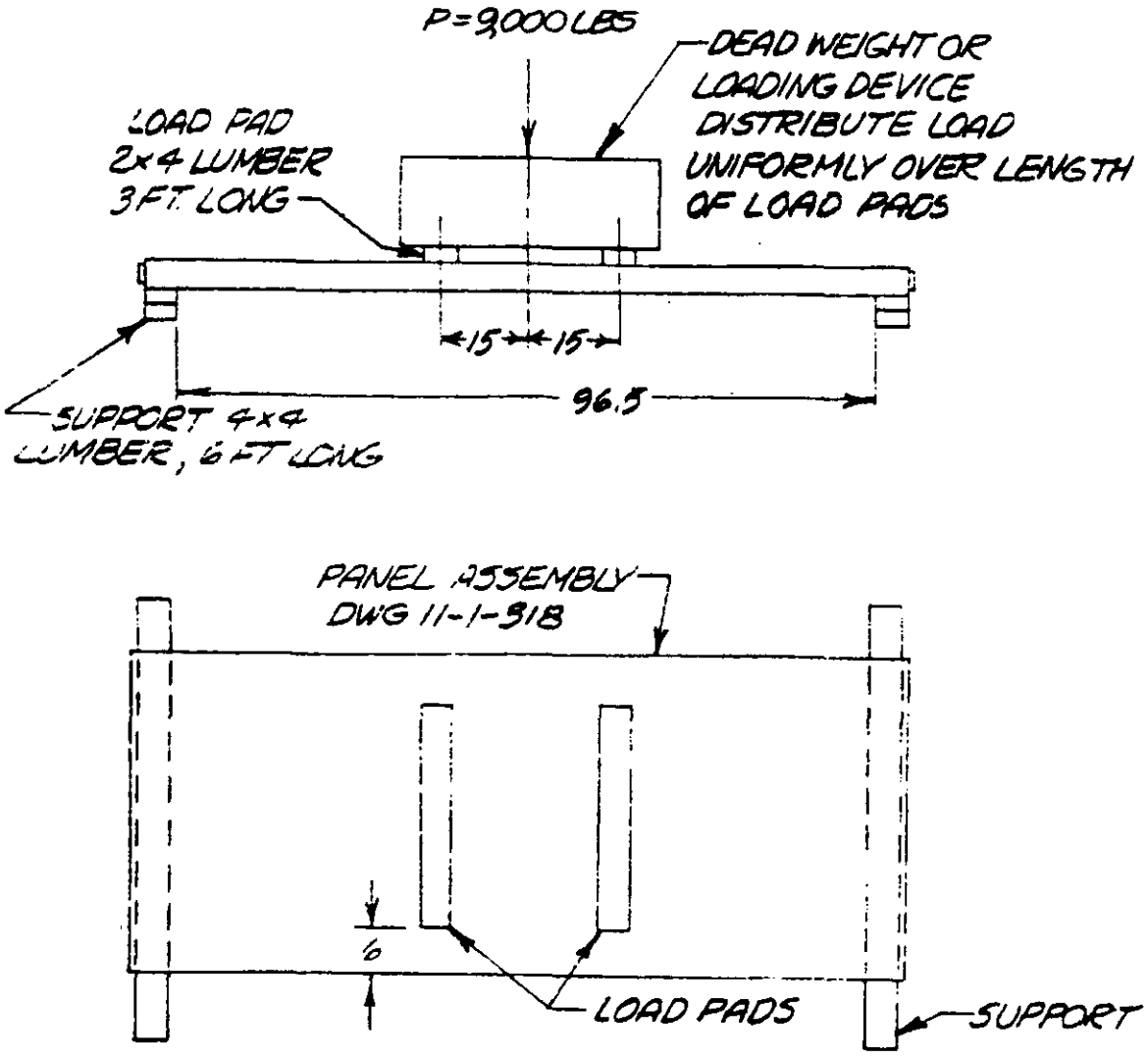


FIGURE 1. Panel proofload test no.1 (dimensions are in inches).

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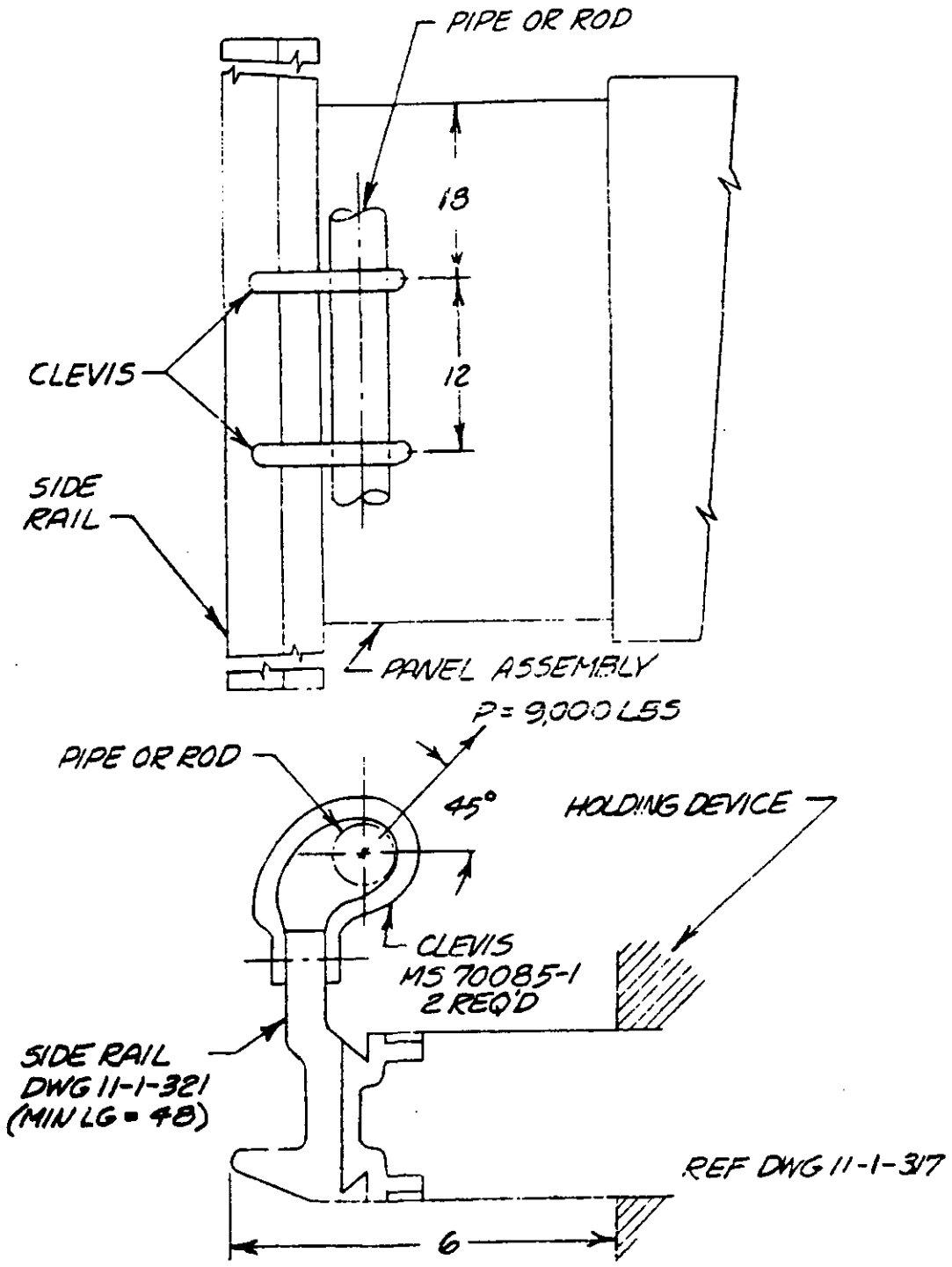


FIGURE 2. Panel proofload test no.2 (dimensions are in inches).

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-43772A	Panel, Type II Modular Platform, Airdrop

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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